BIO12023 - Bioinformática Geral - Turma U (2026/2)

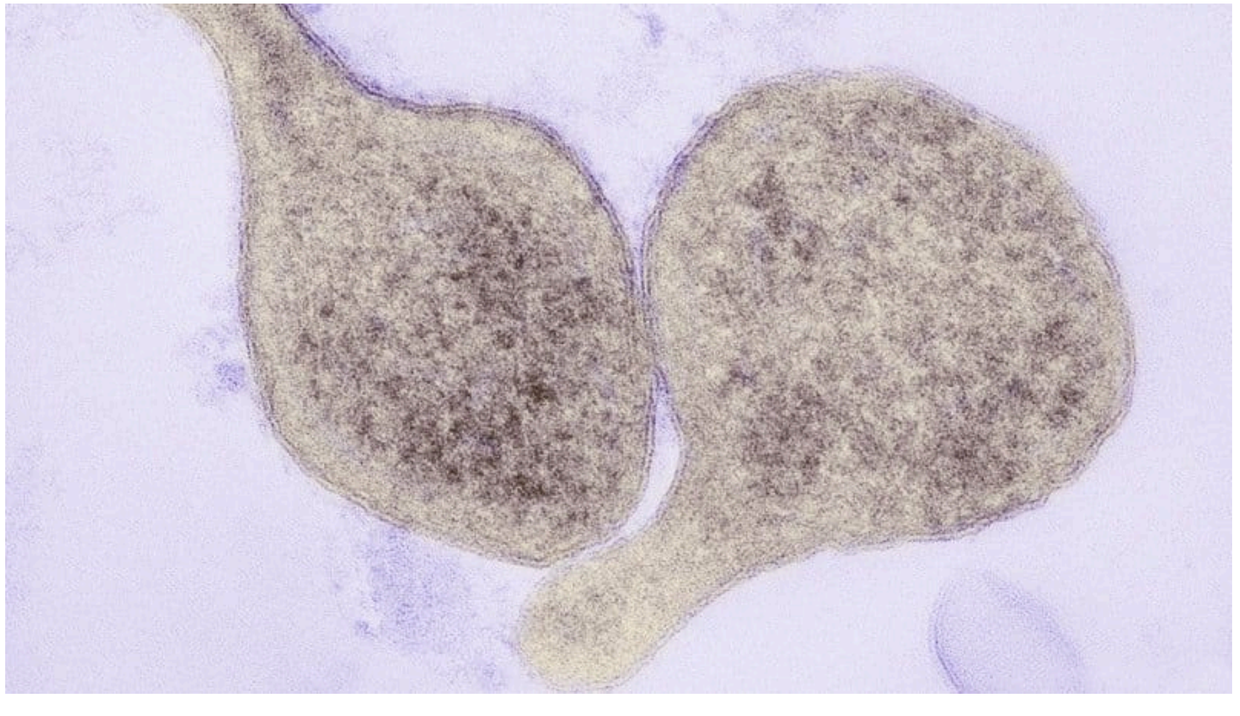
Tarefa 1

Nathalia Dall Agnol Schneider
Gabriel Stracke Oliveira

Professor Mauricio Menegatti Rigo



Mycoplasma genitalium



Cromossomo circular de DNA dupla-fita, com aproximadamente 580 kb.

Full

SRX34638640: Amplicon NGS of Mycoplasma genitalium: Urine and swabs
1 ILLUMINA (Illumina MiSeq) run: 54,426 spots, 10.9M bases, 4.7Mb downloads

Design: PCR + Illumina DNAPrep

Submitted by: Vall d'Hebron Institut de Recerca

Study: Mycoplasma genitalium in Barcelona, Spain
PRJNA1506641 • SRP723648 • All experiments • All runs

Sample:
SAMN62141217 • SRS30204095 • All experiments • All runs
Organism: Mycoplasmoides genitalium

Library:
Name: MG122
Instrument: Illumina MiSeq
Strategy: AMPLICON
Source: GENOMIC
Selection: PCR
Layout: PAIRED

Runs: 1 run, 54,426 spots, 10.9M bases, 4.7Mb

Run	# of Spots	# of Bases	Size	Published
SRR39974648	54,426	10.9M	4.7Mb	2026-08-03

ID: 46023957

Genome

Selected taxa

Mycoplasmoides genitalium Enter one or more taxonomic names

Filters

Download Select columns 40 Genomes Rows per page 20 1-20 of 40

Assembly	GenBank	RefSeq	Scientific name	Modifier	Action
ASM4055692v1	GCA_040556925.1	GCF_040556925.1	Mycoplasmoides genitalium	G37 (strain)	
ASM2732v1	GCA_000027325.1	GCF_000027325.1	Mycoplasmoides genitalium G37	G-37 (strain)	
ASM5492645v1	GCA_054926455.1	GCF_054926455.1	Mycoplasmoides genitalium	MGA663 (strain)	
ASM5492659v1	GCA_054926595.1	GCF_054926595.1	Mycoplasmoides genitalium	MGA92 (strain)	
ASM5492639v1	GCA_054926395.1	GCF_054926395.1	Mycoplasmoides genitalium	MGA719 (strain)	
ASM5492651v1	GCA_054926515.1	GCF_054926515.1	Mycoplasmoides genitalium	MGA600 (strain)	
ASM5492649v1	GCA_054926495.1	GCF_054926495.1	Mycoplasmoides genitalium	MGA649 (strain)	
ASM5492656v1	GCA_054926565.1	GCF_054926565.1	Mycoplasmoides genitalium	MGA122 (strain)	
ASM5650089v1	GCA_056500895.1	GCF_056500895.1	Mycoplasmoides genitalium	MGA1109 (strain)	
ASM5492663v1	GCA_054926635.1	GCF_054926635.1	Mycoplasmoides genitalium	MGA38 (strain)	
ASM5277828v1	GCA_052778285.1	GCF_052778285.1	Mycoplasmoides genitalium	MGA20 (strain)	

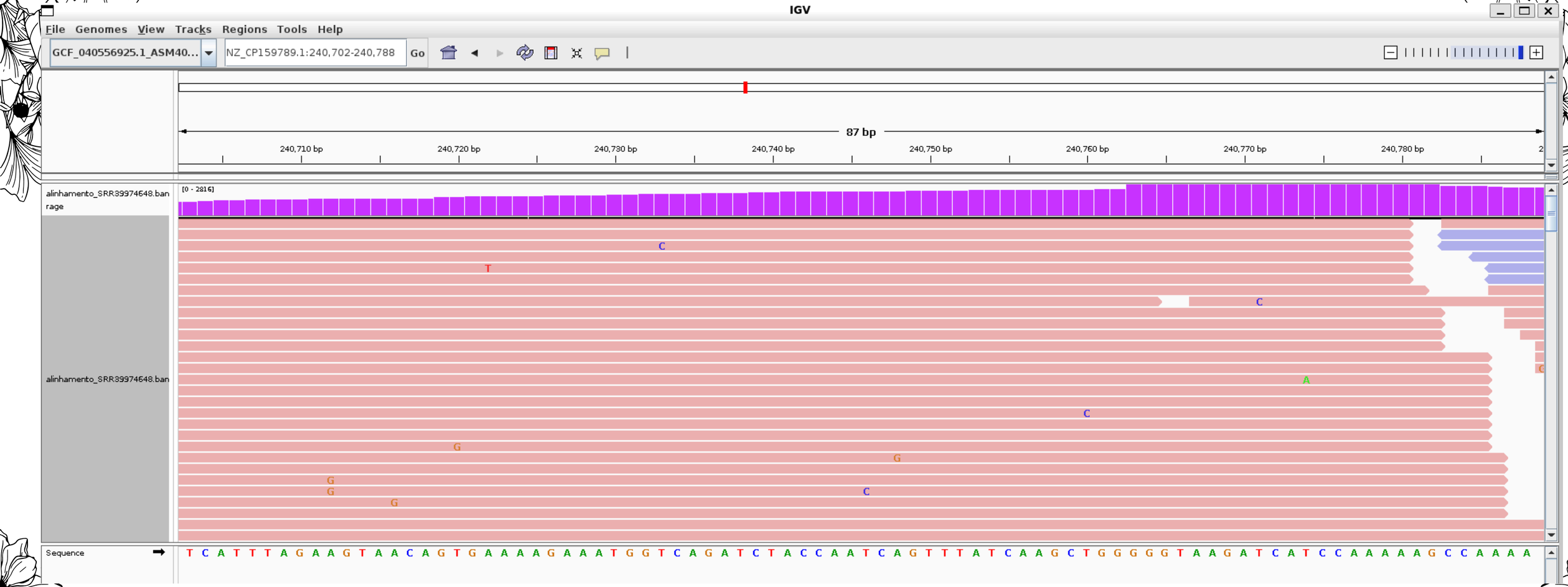
Genome assembly ASM4055692v1 reference

Download datasets API FTP

		Actions
NCBI RefSeq assembly	GCF_040556925.1	
Submitted GenBank assembly	GCA_040556925.1	
Taxon	Mycoplasmoides genitalium	
Strain	G37	
Relation to type material	assembly from type material	
Submitter	Vircell S.L.	
Date	Jul 8, 2024	

View annotated genes

O NCBI remove sequências contaminantes, padroniza a nomenclatura dos cromossomos/contigs e fornece anotações gênicas estruturadas e revisadas.


IGV
File Genomes View Tracks Regions Tools Help
GCF_040556925.1_ASM40...
NZ_CP159789.1:240,702-240,788
Go
87 bp
240,710 bp
240,720 bp
240,730 bp
240,740 bp
240,750 bp
240,760 bp
240,770 bp
240,780 bp
alinhamento_SRR39974648.bam
rage
[0 - 2816]
alinhamento_SRR39974648.bam
Sequence
T C A T T T A G A A G T A A C A G T G A A A A G A A A T G G T C A G A T C T A C C A A T C A G T T T A T C A A G C T G G G G G T A A G A T C A T C C A A A A A G C C A A A A

Os sequenciadores de nova geração (como Illumina) perdem precisão e acumulam mais erros de leitura justamente nas pontas/extremidades das reads.



Download e conversão das reads

```
prefetch SRR39974648
fasterq-dump --split-files SRR39974648
```

Indexação da referência

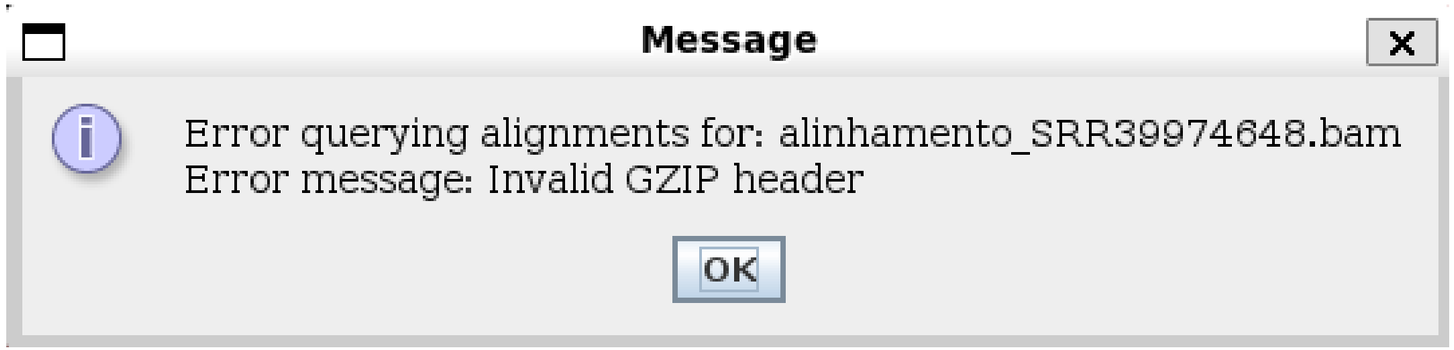
```
minimap2 -d M_genitalium.mmi GCF_040556925.1_ASM4055692v1_genomic.fna
```

Alinhamento

```
minimap2 -ax sr M_genitalium.mmi SRR39974648_1.fastq SRR39974648_2.fastq > alinhamento_SRR39974648.sam
```

Ordenação e indexação do BAM

```
samtools sort -o alinhamento_SRR39974648.bam alinhamento_SRR39974648.sam
samtools index alinhamento_SRR39974648.bam
```




```
picard SortSam \
I=alinhamento_SRR39974648.sam \
O=alinhamento_SRR39974648.bam \
SORT_ORDER=coordinate \
CREATE_INDEX=true
```